Classical Mechanics I (Spring 2026): Homework #1 Solution

Due Mar. 23, 2026

[0.5 pt each, total 6 pts]

1. Thornton & Marion, Problem 2-5

(Note: For Problems 2-5, assume that the pilot intends to keep his Mach 3 speed during the circular maneuver.)

• (b) The minimum radius is attained when the pilot experiences $9g$ at the bottom of the circular maneuver. The net centripetal force there is then $F = m \cdot (9g - g) = \frac{mv^2}{R_{\min}}$.

2. Thornton & Marion, Problem 2-15

(Note: For Problem 2-15, you may need to utilize an integral table; see the fourth bullet point above.)

• $m\dot{v} = mg\sin\theta - kmv^2 \quad \rightarrow \quad \frac{dv}{v^2 - (g/k)\sin\theta} = -kdt \quad \rightarrow \quad \tanh^{-1}\left(\frac{v}{\sqrt{(g/k)\sin\theta}}\right) = \sqrt{gk\sin\theta} \cdot t$ where we utilized Eq.(E.4c) in Appendix E.

3. Thornton & Marion, Problem 2-25

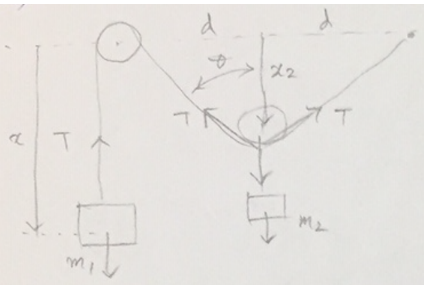
• (a) normal force $N = mg + \frac{mv^2}{R} = mg + \frac{2mgh}{R}$

• (b) $N = mg\cos 45° + \frac{mv'^2}{R} = \frac{mg}{\sqrt{2}} + \frac{m}{R}\left[2gh - 2gR\left(1 - \frac{1}{\sqrt{2}}\right)\right]$

4. Thornton & Marion, Problem 2-44

(Note: For Problem 2-44, derive step by step the equation of motion depicting the small oscillation; this problem was briefly discussed in the class.)

• Please see the scanned image attached.



$$\text{EoM}\begin{cases} m_1\ddot{x} = m_1 g - T \\ m_2\ddot{x}_2 = m_2 g - 2T\cos\theta \end{cases}$$

$\cos\theta = \dfrac{2x_2}{b-x}$

$$\hookrightarrow m_2\ddot{x}_2 = m_2 g - 2(m_1 g - m_1\ddot{x})\cos\theta$$

$$\hookrightarrow \frac{m_2(\ddot{x}_2 - g)}{4x_2} = \frac{m_1(\ddot{x}-g)}{(b-x)} \;//$$

If $\ddot{x} = \ddot{x}_2 = 0$,

$$\frac{m_2 g}{4x_2} = \frac{m_1 g}{b-x}$$

$$\rightarrow m_2(b-x) = 4m_1 x_2 = 4m_1\sqrt{\left(\frac{b-x}{2}\right)^2 - d^2}$$

↳ same as previous p.

$$\hookrightarrow \begin{cases} x_0 = b - \dfrac{4m_1 d}{\sqrt{4m_1^2 - m_2^2}} \\ x_{2,0} = \dfrac{m_2 d}{\sqrt{4m_1^2 - m_2^2}} \end{cases}$$

$x = x_0 + r$, $x_2 = x_{2,0} + r_2$

$$\text{EoM} \rightarrow \frac{m_2(\ddot{r}_2 - g)}{4(x_{2,0}+r_2)} = \frac{m_1(\ddot{r}-g)}{b-x_0-r}$$

$$\rightarrow \frac{m_2(\ddot{r}_2-g)}{4x_{2,0}}\left(1 - \frac{r_2}{x_{2,0}}\right) \simeq \frac{m_1(\ddot{r}-g)}{b-x_0}\left(1 + \frac{r}{b-x_0}\right)$$

$$\boxed{\begin{aligned}(x_{2,0}+r_2)^{-1} &= x_{2,0}^{-1}\left(1+\frac{r_2}{x_{2,0}}\right)^{-1} \\ &\simeq \frac{1}{x_{2,0}}\left(1 - \frac{r_2}{x_{2,0}}\right)\end{aligned}}$$

$$\rightarrow -\frac{m_1}{4x_{2,0}}\ddot{r} - \cancel{\frac{m_2}{4x_{2,0}}g} - \cancel{\frac{m_1\ddot{r}r_2}{4x_{2,0}^2}} + \frac{-m_1 g r}{4x_{2,0}^2}$$
$$- \frac{m_1\ddot{r}}{b-x_0} + \cancel{\frac{m_1 g}{b-x_0}} - \cancel{\frac{m_1 r\ddot{r}}{(b-x_0)^2}} + \frac{m_1 g r}{(b-x_0)^2} = 0$$

$$\hookrightarrow \left(\frac{m_1}{4x_{2,0}} + \frac{m_1}{b-x_0}\right)\ddot{r} + \left(\frac{m_1 g}{4x_{2,0}^2} - \frac{m_1 g}{(b-x_0)^2}\right)r = 0$$

$$\hookrightarrow \ddot{r} + \underbrace{\frac{g(4m_1^2 - m_2^2)^{\frac{3}{2}}}{4m_1 m_2(m_1+m_2)d}}_{\omega^2} r = 0$$

$$\begin{cases} d^2 + x_2^2 = \left(\dfrac{b-x}{2}\right)^2 \rightarrow \cancel{d^2} + x_{2,0}^2\left(\cancel{1} + \dfrac{2r_2}{x_{2,0}}\right) \simeq \left(\dfrac{b-x_0}{2}\right)^2 \cdot \left(\cancel{1} - \dfrac{2r}{b-x_0}\right) \\ d^2 + x_{2,0}^2 = \left(\dfrac{b-x_0}{2}\right)^2 \end{cases}$$

$$\downarrow$$

$$x_{2,0}\cdot 2r_2 \simeq \frac{b-x_0}{4}(-2r)$$

$$\downarrow$$

$$\frac{r_2}{r} \simeq \frac{-(b-x_0)}{4x_{2,0}} = \frac{-(4m_1 d)}{4m_2 d} = -\frac{m_1}{m_2} \rightarrow \frac{\ddot{r}_2}{\ddot{r}} = -\frac{m_1}{m_2}$$

5. Thornton & Marion, Problem 2-52

(Note: For Problem 2-52(b), you should feel free to use any numerical tool of your choice, such as MATLAB, MATHEMATICA, or simply WOLFRAMALPHA.)

- (b) $F(x) = -\frac{4U_0 x}{a^2}\left(1 - \frac{x^2}{a^2}\right) = 0$ at $x = 0, \pm a$
- (c) At $x = 0$, $m\ddot{x} \approx -\frac{4U_0 x}{a^2} \quad \rightarrow \quad \omega_0^2 = \frac{4U_0}{ma^2}$
- (d) $\frac{1}{2}mv_{\text{min}}^2 + U(x=0) = U(x=\pm a) \quad \rightarrow \quad v_{\text{min}} = \sqrt{\frac{2U_0}{m}}$

6. Thornton & Marion, Problem 3-2

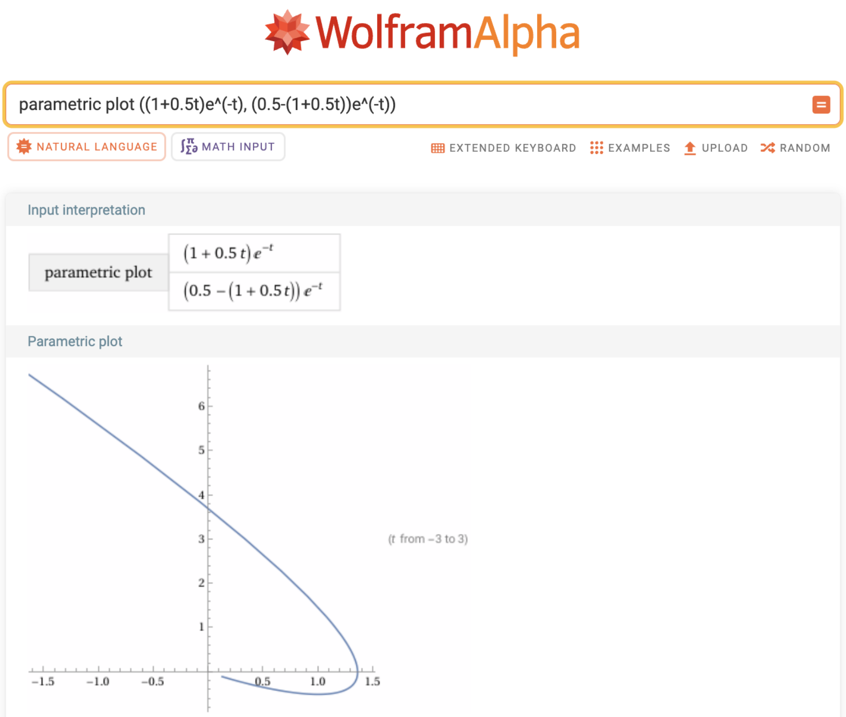
(Note: For Problem 3-2, the decrement of the motion is defined in Eq.(3.42).)

- From Problem 3-1 the natural frequency of the undamped oscillator is $\omega_0 = 10\,\text{s}^{-1}$, while the frequency of the damped oscillator is $\omega_1 = \sqrt{\omega_0^2 - \beta^2}$.

7. Thornton & Marion, Problem 3-21

(Note: For Problem 3-21 and all other numerical problems, you are asked to submit your source code — or a proof of your work — as well; see the second bullet point above.)

- $x = x_0 + (v_0 + \beta x_o)te^{-\beta t}, \quad v = [v_0 - \beta(v_0 + \beta x_o)t]e^{-\beta t}$

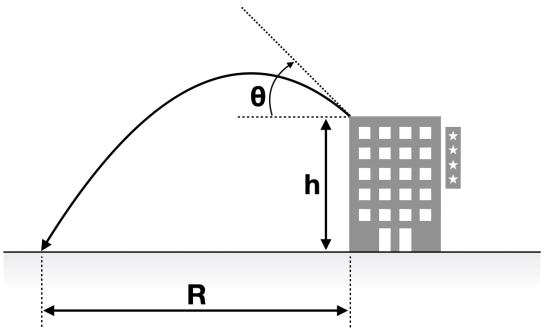


8. Thornton & Marion, Problem 3-26

- $m_1\ddot{x}_1 + b_1(\dot{x}_1 - \dot{x}_2) + kx_1 = F\cos\omega t, \quad m_2\ddot{x}_2 + b_1(\dot{x}_2 - \dot{x}_1) + b_2\dot{x}_2 = 0$

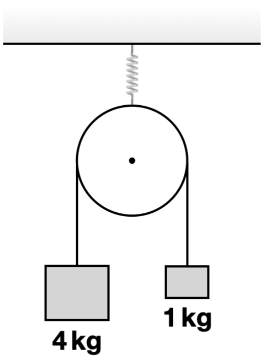
- $L_1\ddot{q}_1 + R_1(\dot{q}_1 - \dot{q}_2) + (1/C)q_1 = E_0\cos\omega t, \quad L_2\ddot{q}_2 + R_1(\dot{q}_2 - \dot{q}_1) + R_2\dot{q}_2 = 0$ if the substitutions seen in Examples 3.4 and 3.5 are applied.

9. A person fires a cannon from the top of a building of height h above a horizontal plain. Show that, for a given muzzle velocity v_i, the maximum horizontal range is achieved when the cannon is fired at an angle θ satisfying $\csc^2\theta = 2\left(1 + \frac{gh}{v_i^2}\right)$. (Note: You may want to review the *implicit* differentiation, which appears in elementary calculus textbooks such as Stewart or 김홍종, or in mathematical physics textbooks like Boas (Chapter 4.6).)



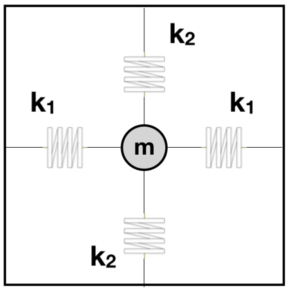
- From $R = (v_i\cos\theta)t_0$ and $0 = h + (v_i\sin\theta)t_0 - \frac{1}{2}gt_0^2$, one obtains an equation relating R and θ, $R^2 - \frac{v_i^2}{g}(\sin 2\theta)R - \frac{2hv_i^2}{g}\cos^2\theta = 0$. Differentiating the equation with respect to θ and setting $\frac{dR}{d\theta} = 0$ yields $R = h\tan 2\theta$. Plugging this back into the previous equation leads to the desired condition on θ.

10. Consider a pulley of mass 2 kg suspended from the ceiling by a spring balance. Two masses of 4 kg and 1 kg are attached to the opposite ends of a light string passing over the pulley, which is frictionless. When the system is released, the masses begin to move under the influence of gravity. While the system is in motion, determine the reading shown by the spring balance. Explain qualitatively why this reading is not simply 7 kg.



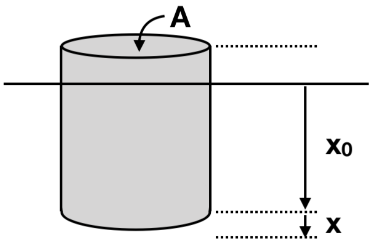
- $m_1 a = m_1 g - T$ and $m_2 a = T - m_2 g$, while the reading of the spring balance is $2T + m_{\text{pulley}}g$.

11. In the class we discussed two-dimensional simple harmonic oscillators. (a) Describe an example of the physical realization of Eqs. (3.19) and (3.27) using the figure shown below. (b) Reproduce Figures 3-2 to 3-4 in your textbook with your favorite numerical tool. For Figure 3-2, you don't have to remake all 10 panels; 2-3 panels should be enough.



• A mass m is tied to four springs, two of constant k_1 in the x-direction, and two others of constant k_2 in the y-direction. One end of each of the springs is tied to m (in a + shape), and the other end is tied to walls. At equilibrium none of the springs is stretched. We consider only small oscillations from the equilibrium point.

12. An object of mass m, density $\rho_1 = 0.75\,\mathrm{g\,cm^{-3}}$, and uniform horizontal cross-sectional area $A = 1\,\mathrm{cm^2}$ is floating in a liquid of density $\rho_2 = 1.0\,\mathrm{g\,cm^{-3}}$. At equilibrium, the object displaces a volume $V_0 = 0.75\,\mathrm{cm^3}$ of the liquid.



(a) When the object is slightly pushed down from its equilibrium position, write down the equation governing small oscillations, while neglecting the retarding force of the liquid. Express the characteristic angular frequency in terms of the variables given, and evaluate its numerical value.

(b) Now consider the retarding force exerted by the liquid, whose magnitude for this object can be written as $2m\sqrt{g/c_\mathrm{r}}\,\dot{x}$, with $c_\mathrm{r} = 245\,\mathrm{cm}$. Write down the equation of small oscillations. Express its solution in terms of the variables given, and evaluate the numerical values of any coefficients that can be determined explicitly. State whether the motion is underdamped, overdamped, or critically damped.

(c) Determine the time required for the oscillation amplitude to fall below e^{-10} times its initial maximum value.

• (a) $m\ddot{x} = mg - \rho_2(V_0 + \Delta V)g \;\; \rightarrow \;\; m\ddot{x} = -\rho_2 \Delta V g = -\rho_2 g A x$ since $0 = mg - \rho_2 V_0 g$. Therefore, $\omega_0 = \sqrt{\frac{\rho_2 g A}{m}} = \sqrt{\frac{gA}{V_0}} \simeq 36.15\,\mathrm{s^{-1}}$.

• (b) $m\ddot{x} = -\rho_2 g A x - 2m\sqrt{\frac{g}{c_\mathrm{r}}}\dot{x} \;\; \rightarrow \;\; \ddot{x} + 2\sqrt{\frac{g}{c_\mathrm{r}}}\dot{x} + \omega_0^2 x = 0$. Thus, the oscillation amplitude is $x(t) = De^{-\sqrt{g/c_\mathrm{r}}t}\cos(\omega_1 t - \delta)$ with $\sqrt{\frac{g}{c_\mathrm{r}}} = 2\,\mathrm{s}^{-1}$ and $\omega_1 = \sqrt{\omega_0^2 - \frac{g}{c_\mathrm{r}}} = \sqrt{g\left(\frac{1\,\mathrm{cm}^2}{0.75\,\mathrm{cm}^3} - \frac{1}{245\,\mathrm{cm}}\right)} \simeq 36.09\,\mathrm{s}^{-1}$.

• (c) The initial maximum amplitude is $x_\mathrm{max} = D\cos\delta$ at $t = 0 \;\; \rightarrow \;\; x_\mathrm{max}e^{-\sqrt{g/c_\mathrm{r}}t} = x_\mathrm{max}e^{-2t} = x_\mathrm{max}e^{-10}$ at $t = 5\,\mathrm{s}$.